

The bill says how much. It cannot prove it is right.

Indian AI API startups bill in rupees for millions of token events a day. What goes wrong upstream:

ONE USAGE EVENT, POSTED TWICE

POST /v1/events

```
"event_id": "evt_retry_10"  
"total_tokens": 50000  
"timestamp": "12:00:00"
```

network retry ->

```
"timestamp": "12:00:02"
```

Upstream keys its event store on the client timestamp, and its Redis dedup lock expires after 24 hours. A retry with a new timestamp is stored as a second row.



Same event.
Billed twice.

RETRY · DOUBLE-BILLED

2x billed for one request

timestamp sits in the ReplacingMergeTree sort key, so FINAL keeps both rows.

SUB-PAISE RATE · LOST REVENUE

Rs 0.002 per token rounds to 0 paise

INR is hard-coded to 2 decimals; per-token charges round away before the invoice.

● Mid-month upgrades mis-prorate

Plan-change v2 cannot change interval; credits fall back to list price when a read fails.

● Invoices explain nothing

A total with no tokens-to-units, tier or proration breakdown is a support ticket.

● 11 verified upstream gaps

Each with path:line evidence at Flexprice 31421e9 in docs/GAPS.md.

One event in. One **exact** bill out.

Paise-Perfect: a clean-room billing engine in Python + FastAPI + SQLite, built from docs/ alone.

CUSTOMER PLAN

- Basic Rs 60 / month
- Growth Rs 120 / month
- Tokens: slab tiers

USAGE EVENT

event_id **evt_retry_10**
tokens **1,500,000**
upgrade **day 15**

Money is integer paise; rates are integer micro-paise (1 paise = 10,000). No float ever touches money.

INGEST

UNIQUE event_id

METER

SUM / 1,000 = units

PRICE + PRORATE

Deterministic rating engine

Slab or volume tiers with inclusive bounds, rated in micro-paise and rounded half-up once. Day-based proration: 15/30 = 0.5000.

BILL · INTEGER PAISE

Rs 260.00 sealed invoice

Rs 120 plan + Rs 105 tier 1 + Rs 35 tier 2. Idempotent per period.

EXPLAIN · "WHY?"

"1,500,000 tokens = 1,500 units..."

Plain-language bill from local facts; AI only rewords it.

ALERT · RUNAWAY AGENT

Spike detected: 10x normal

Last hour vs 7-day hourly average, flagged above 3x.

Not just a rebuild. **Money that is never a float.**

Three tests, written **before** the engine.

Every expected value is quoted from docs/PRD.md section 5. Committed red, then built until green.

KILLER TEST 1

PASS

Duplicate event counts once

50,000 tokens / 1,000	50 units
retry, changed timestamp	200 skipped
8 concurrent threads	1 row
invoice line quantity	50, not 100

KILLER TEST 2

PASS

Day-15 upgrade prorates exactly

30-day April cycle	upgrade Apr 16
coefficient	15/30 = 0.5000
unused Starter credit	-150,000 p
remaining Pro charge	+450,000 p
net amount due	300,000 p

KILLER TEST 3

PASS

Tiered pricing to the paisa

1,500,000 tokens	1,500 units
slab tier 1: 1,000 x 10p + Rs 5	10,500 p
slab tier 2: 500 x 5p + Rs 10	3,500 p
slab total	Rs 140.00
volume: 1,500 x 5p + Rs 10	Rs 85.00

pytest: 44 passed **16 Killer · 10 Fix · 14 Differentiator · 4 Dashboard**

Edge cases pinned too: exact tier boundary stays in Tier 1; Day-1 and Day-30 upgrades; aggregate-then-tier.

Rules compute. AI **only** explains.

FIX · IMPROVEMENT 1

CLOSES GAPS 1, 7, 11

Deterministic dedup + sub-paise rating

UNIQUE(tenant_id, event_id) in SQLite: a retry with a new timestamp, or a batch replayed days later, still dedupes. No cache TTL to outlive.

Rated in micro-paise, rounded once: 1,500,500 tokens bill 14,003 paise, not 14,005; Rs 0.002/token over 10 requests bills 2 paise, not 0.

INGEST OUTCOMES

201 CREATED

200 DUPLICATE

409 SEALED PERIOD

400 NOT WHOLE

A late event can never land in a finalized invoice.

DIFFERENTIATOR · IMPROVEMENT 2

Invoice explainer + spike alert

GET /v1/invoices/{id}/explanation · GET /v1/customers/{id}/spike-status. Closes gaps 8 and 10.

"EXPLAIN MY BILL"

LIVE IN /app

Customer: "Why is my bill Rs 260?"

Engine: "You used 1,500,000 tokens (1,500 units). Tier 1: 1,000 units at 10 paise = Rs 105. Tier 2: 500 units at 5 paise = Rs 35. Plus the Rs 120 plan fee."

Every figure is computed locally. No AI key or an AI failure: the template still answers, HTTP 200.

WHY OURS IS DIFFERENT

Versus the upstream engine

1. Storage-level dedup

A database constraint, not a 24h cache.

2. Integer money

Paise and micro-paise; one rounding step.

3. Injectable clock

BILLING_NOW tests a month in milliseconds.

4. Sealed periods stay sealed

409 for late events into a final invoice.

5. Explainable bills

Plain words, never invented numbers.

6. Runaway-agent alert

Flags a prompt loop before the invoice.

Run it locally. Or open it live.

44

pytest cases
all passing

67

demo checks
every one PASS

~2s

suite runtime
BILLING_NOW

0

external services
SQLite only

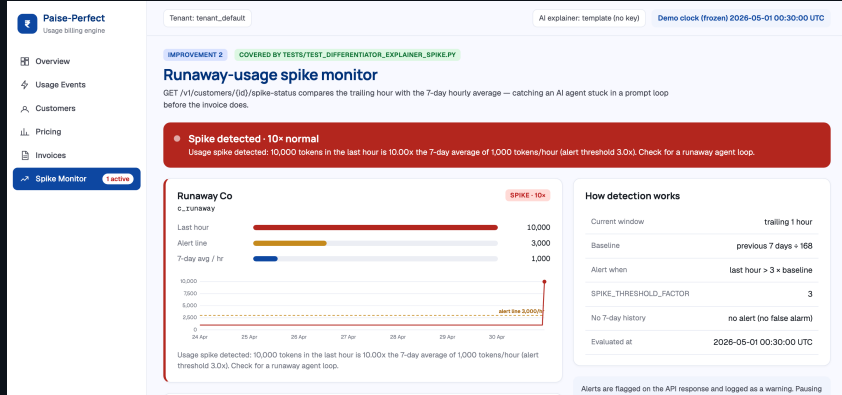
RUN IT

```
pip install -r requirements.txt
cp .env.example .env
pytest -q                # 44 passed
./scripts/demo            # 67 PASS
./scripts/ui              # /app dashboard
Windows: scripts\demo, scripts\ui
```

VERIFIED LIVE

- ✓ Duplicate + 8-way concurrent events count once
- ✓ Day-15 upgrade nets exactly Rs 3,000.00
- ✓ Slab Rs 140 / Volume Rs 85, to the paisa
- ✓ Rs 0.002/token bills 2 paise, not 0
- ✓ Explanation returns 200 with no AI key
- ✓ Runaway loop flagged at 10x baseline

LIVE DASHBOARD · /APP



Live preview: paise-perfect.vercel.app · github.com/Pranav-422/Reverse-Flexprice